PHASE CHANGES

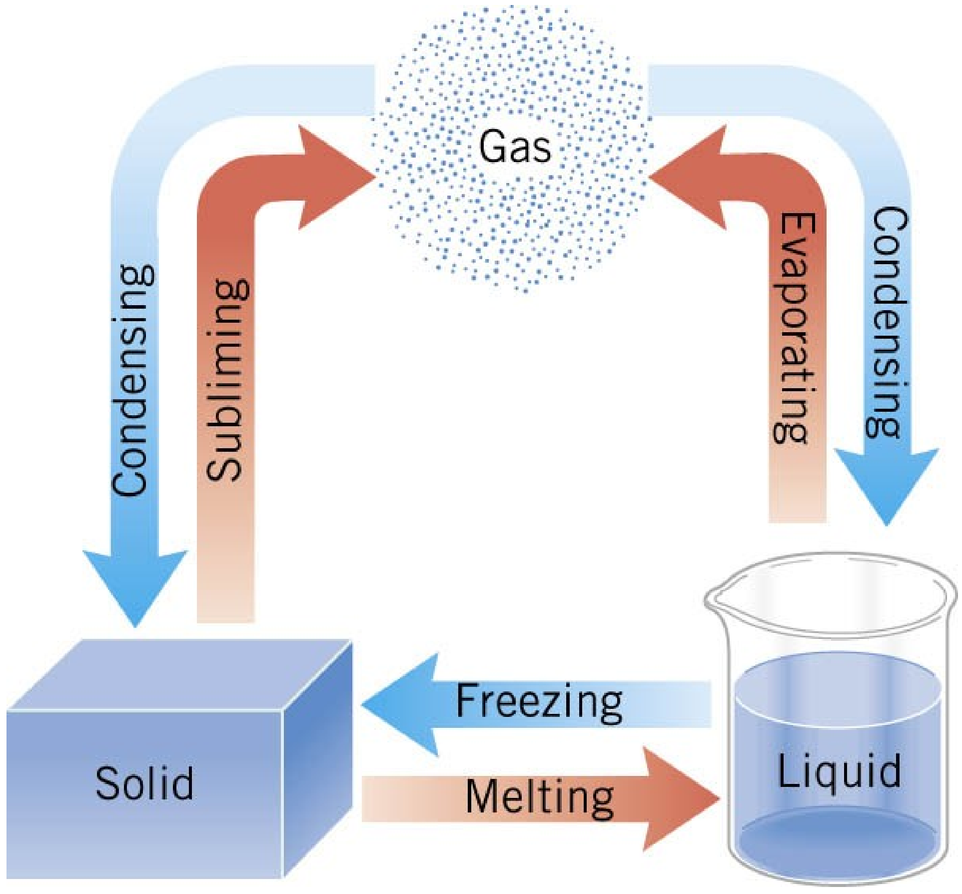
References: - Physics / Johnson and Cutnell / Wiley (slides)
- Conceptual Physics / Paul Hewitt / Pearson
- Physics for future presidents / Dr. Richard Muller / Princeton

STATES OF MATTER

The ancients describe the state of matter as:
Earth, water, air and fire .

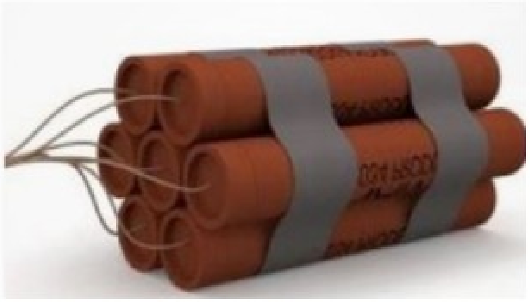
Is it stupid ?

THE PHASES OF MATTER



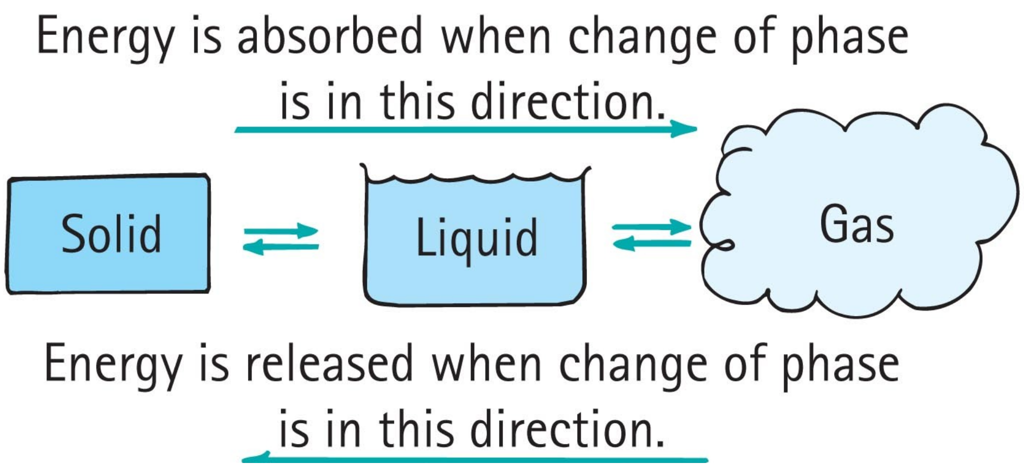
https://phet.colorado.edu/sims/html/states-of-matter-basics/latest/states-of-matter-basics_en.html

SOLID GOES TO GAS, volume is multiplied by 1000
If the gas is hot, volume is multiplied by 100,000



Energy and Change of Phase

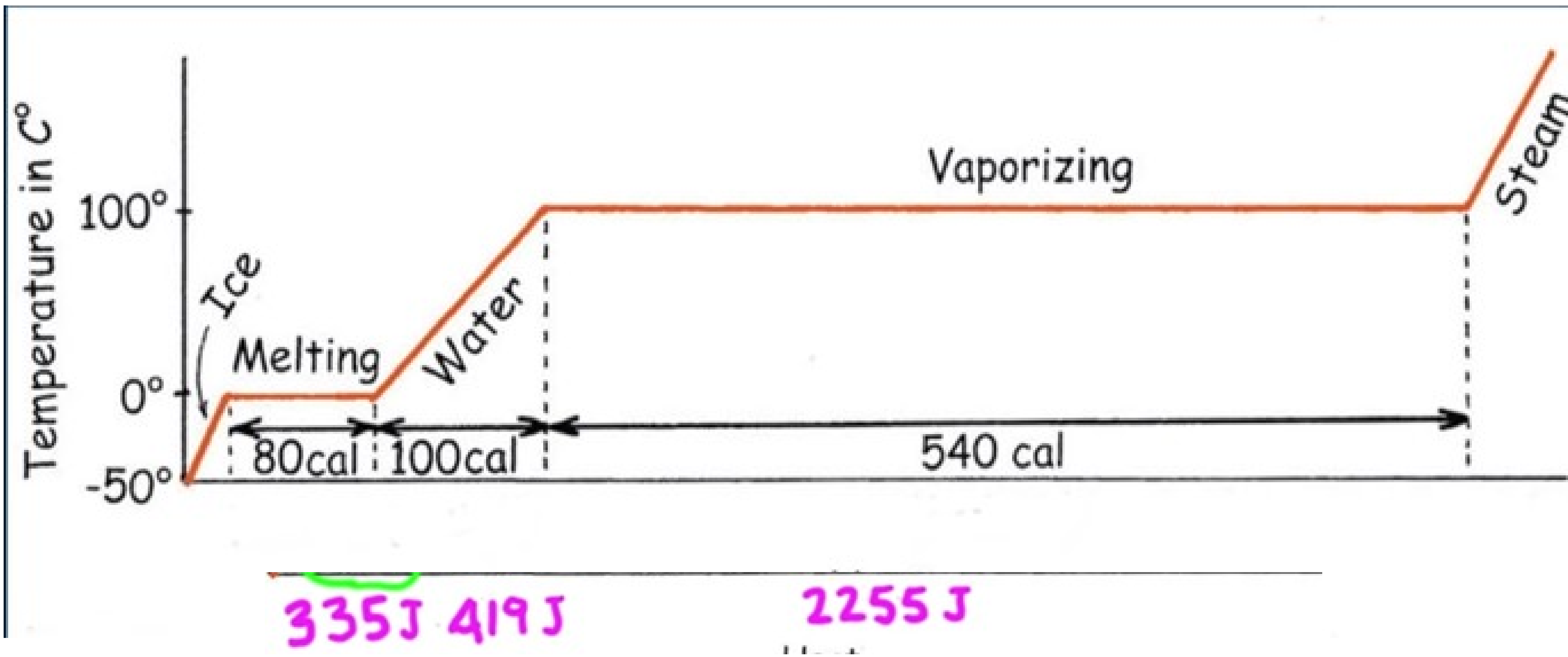
- Energy and Change of Phase
  - From solid to liquid to gas phase
    - add energy
  - From gas to liquid to solid phase
    - remove energy



Phases of Matter

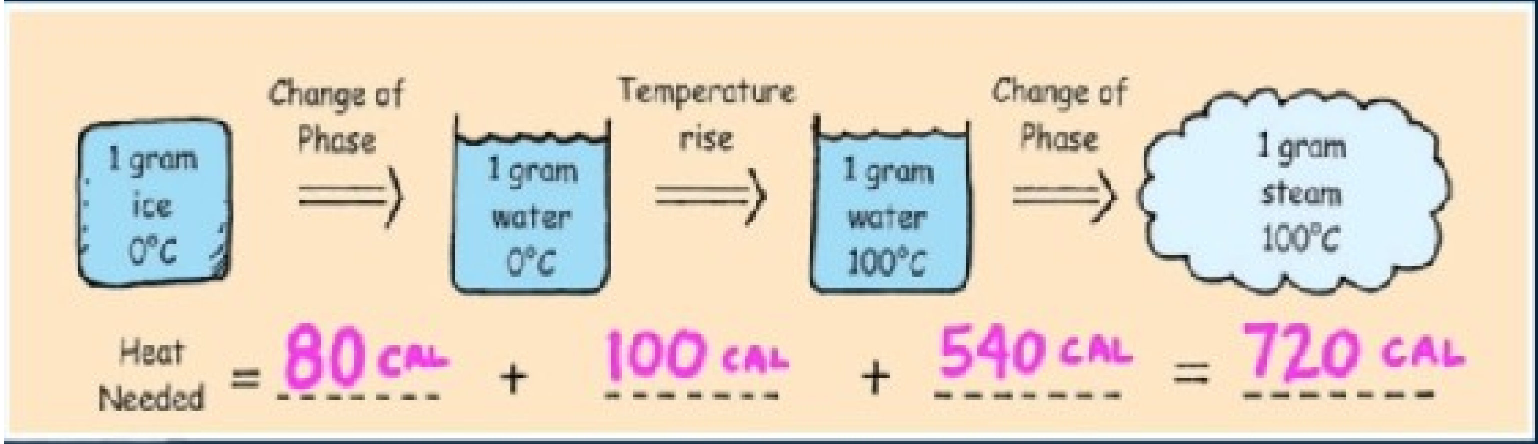
- The phase of material depends upon the temperature and pressure.
- Change from Solid $\rightarrow$ Liquid $\rightarrow$ Gas $\rightarrow$ Plasma requires energy to be added to the material.
- Energy causes the molecules to move more rapidly.

During a phase change, the temperature of the mixture does not change (provided the system is in thermal equilibrium).
It takes a lot of energy to turn liquid water to vapor.
So vapor has a huge amount of energy
(used for radiator, for heat engine, vapor burn is worse than hot water)



See the slopes ? It takes 1 cal to increase the temperature of water by 1 degree (its a lot) so 100 cal to get to 100 C (boiling)
It only takes 0.5 cal to increase the temperature of ice by 1C.

$$Q = mL_f + cm\Delta T + mL_v = 80\text{ CAL} + 100\text{ CAL} + 540\text{ CAL} = 720\text{ CAL}$$



Conceptual Example 13 **Saving Energy**

Suppose you are cooking spaghetti for dinner, and the instructions say “boil pasta in water for 10 minutes.” To cook spaghetti in an open pot with the least amount of energy, should you turn up the burner to its fullest so the water vigorously boils, or should you turn down the burner so the water barely boils?

HEAT SUPPLIED OR REMOVED IN CHANGING THE PHASE OF A SUBSTANCE

The heat that must be supplied or removed to change the phase of a mass *m* of a substance is

$$Q = m\underbrace{L}_{\text{latent heat}}$$

SI Units of Latent Heat: J/kg

12.8 Heat and Phase Change: Latent Heat

Table 12.3 Latent Heats[a] of Fusion and Vaporization

Substance	Melting Point (°C)	Latent Heat of Fusion, L_f (J/kg)	Boiling Point (°C)	Latent Heat of Vaporization, L_v (J/kg)
Ammonia	−77.8	33.2×10^4	−33.4	13.7×10^5
Benzene	5.5	12.6×10^4	80.1	3.94×10^5
Copper	1083	20.7×10^4	2566	47.3×10^5
Ethyl alcohol	−114.4	10.8×10^4	78.3	8.55×10^5
Gold	1063	6.28×10^4	2808	17.2×10^5
Lead	327.3	2.32×10^4	1750	8.59×10^5
Mercury	−38.9	1.14×10^4	356.6	2.96×10^5
Nitrogen	−210.0	2.57×10^4	−195.8	2.00×10^5
Oxygen	−218.8	1.39×10^4	−183.0	2.13×10^5
Water	0.0	33.5×10^4	100.0	22.6×10^5

[a]The values pertain to 1 atm pressure.

Example 14 Ice-cold Lemonade

Ice at 0°C is placed in a Styrofoam cup containing 0.32 kg of lemonade at 27°C. The specific heat capacity of lemonade is virtually the same as that of water. After the ice and lemonade reach and equilibrium temperature, some ice still remains. Assume that mass of the cup is so small that it absorbs a negligible amount of heat.
What is the mass of ice remaining?

Hint: so the final temperature is 0C since some ice remains

12.8 Heat and Phase Change: Latent Heat

$$\underbrace{\left(mL_f\right)_{\text{ice}}}_{\substack{\text{Heat gained}\\ \text{by ice}}} = \underbrace{\left(cm\Delta T\right)_{\text{lemonade}}}_{\substack{\text{Heat lost by}\\ \text{lemonade}}}$$

$$m_{\text{ice}} = \frac{\left(cm\,\Delta T\right)_{\text{lemonade}}}{L_f}$$

$$¿\frac{\left[4186\ J/\left(\text{kg}\cdot C^{\circ}\right)\right]\left(0.32\ \text{kg}\right)\left(27^{\circ}C - 0^{\circ}C\right)}{3.35\times 10^{5}\ J/\text{kg}} = 0.11\ \text{kg}$$

Example 15 Getting Ready for a Party

A 7.00-kg glass bowl [$c = 840$ J/(kg · C°)] contains 16.0 kg of punch at 25.0 °C. Two-and-a-half kilograms of ice [$c = 2.00 \times 10^3$ J/(kg · C°)] are added to the punch. The ice has an initial temperature of −20.0 °C, having been kept in a very cold freezer. The punch may be treated as if it were water [$c = 4186$ J/(kg · C°)], and it may be assumed that there is no heat flow between the punch bowl and the external environment. The latent heat of fusion for water is 3.35×10^5 J/kg. When thermal equilibrium is reached, all the ice has melted, and the final temperature of the mixture is above 0 °C. Determine this temperature.

Nuclear Meltdown!

Reactor power: 200 MW   Reactor fuel: 250 tonnes UO_2

Emergency cooling water: 4000 tonnes (4 million kg)

How long to boil away?

How long for fuel to melt?

Substance	Melt (K)	L_f (kJ/kg)	Boil (K)	L_v (kJ/kg)
Water	273	334	373	2257
UO_2	3120	259	3815	1533

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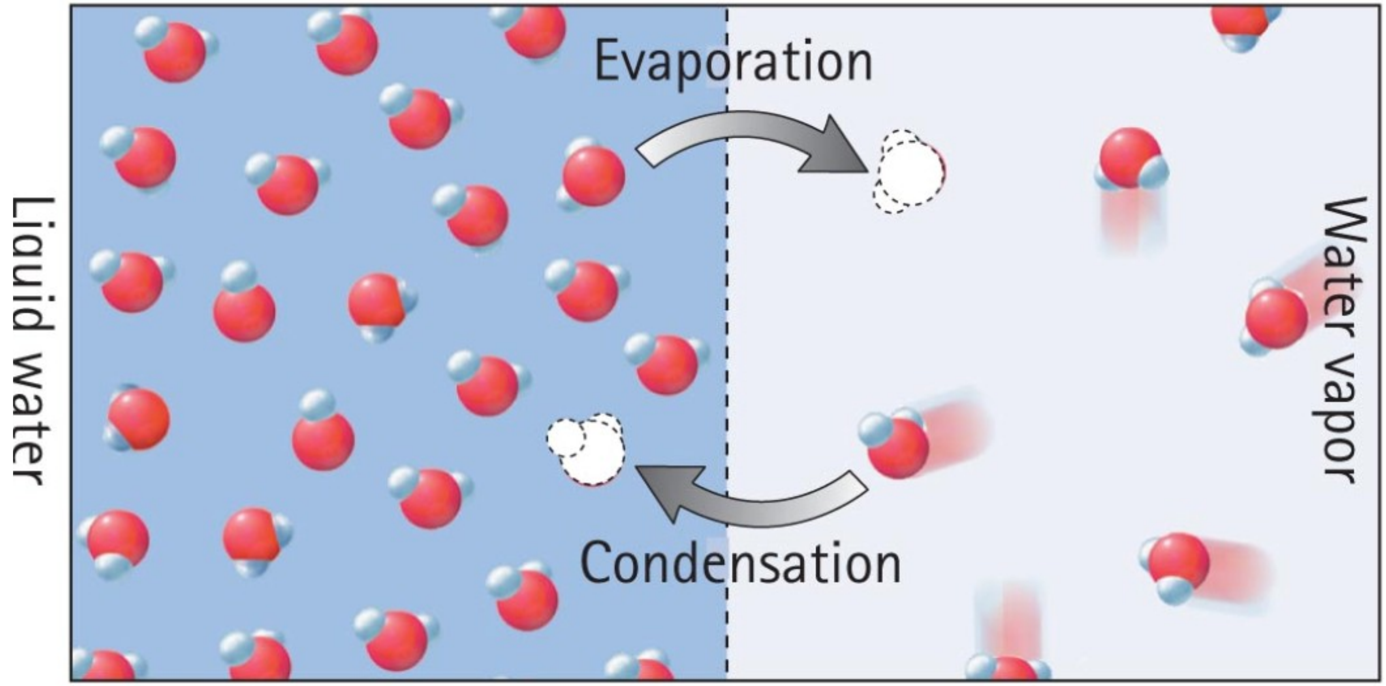
Energy to vaporize water: $Q = L_v m_{water}$ = (2257 kJ/kg)(4×10^6 kg) = 9×10^9 kJ

Reactor power: 200 MW = 200,000 kW = 200,000 kJ/s

Time to boil: (9×10^9 kJ)/(200,000 kJ/s) = 45,000 s = 12.5 hours

Evaporation

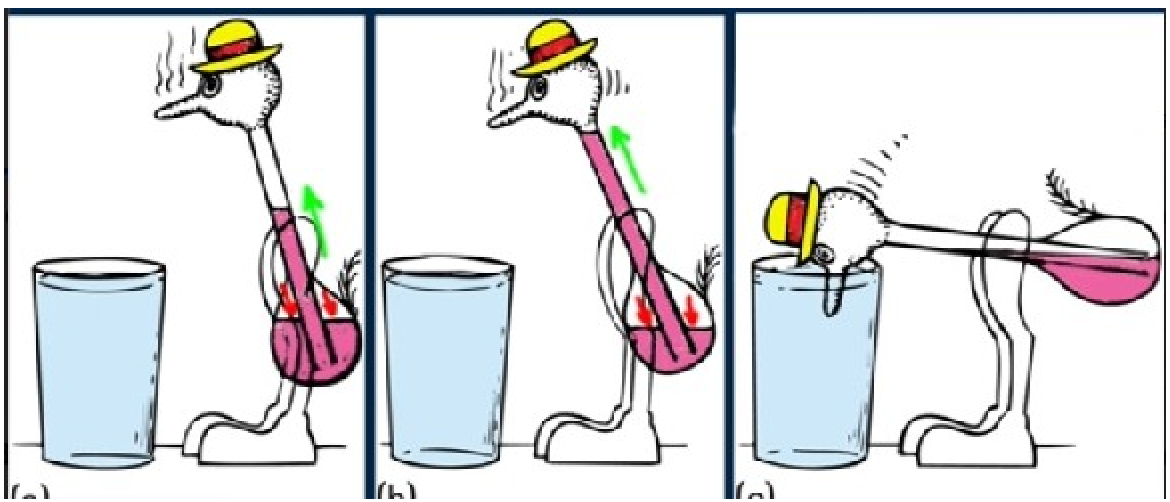
- Evaporation
  - Change of phase from liquid to gas



Condensation is a warming process

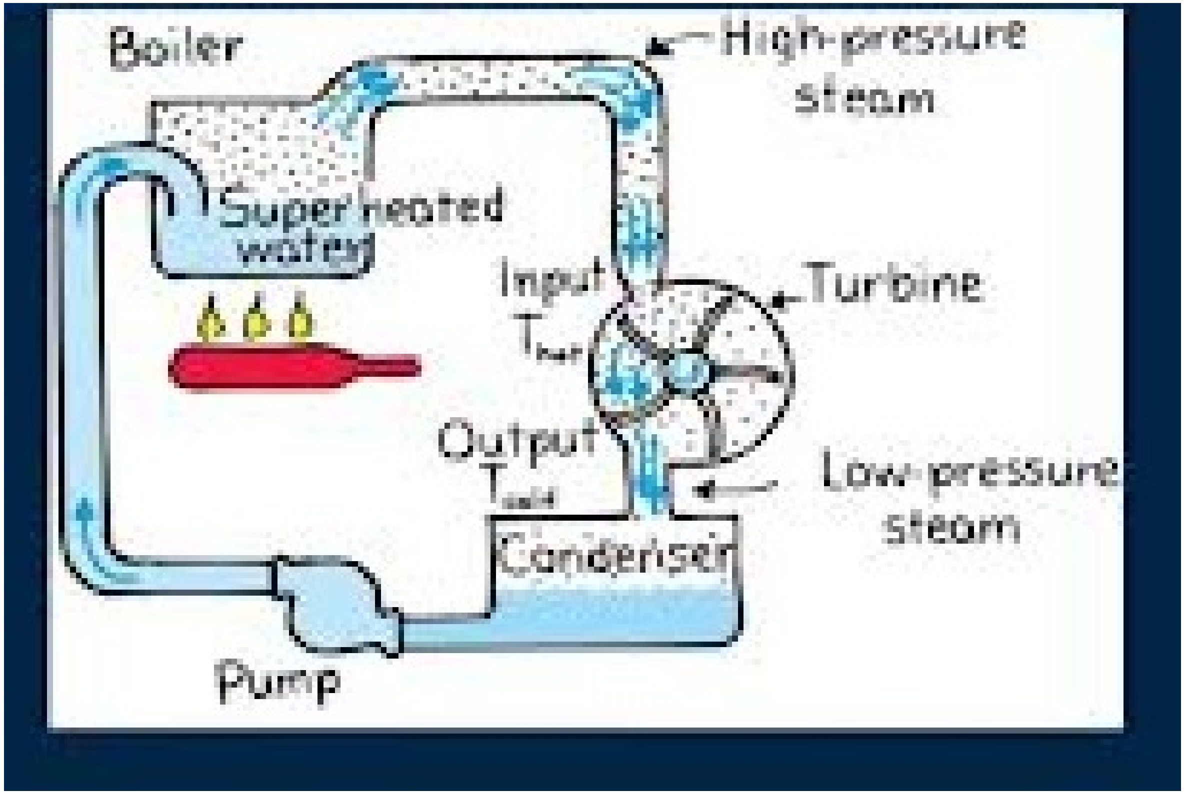
- Condensation process (continued)
  - Kinetic energy is absorbed by the liquid, resulting in increased temperature.
  - Examples:
    - Steam releases much energy when it condenses to a liquid and moistens the skin—hence, it produces a more damaging burn than from same-temperature 100°C boiling water.
    - You feel warmer in a moist shower stall because the rate of condensation exceeds the rate of evaporation.

Lots of energy in water vapor that can be released in
A room through condensation. Condensation is a
Warming process. Burn from steam happens
Because of condensation on a skin.



Ref. Paul Hewitt

The beak is wet. Evaporation
Is a cooling process. The
Inside of the head cools.
There is condensation inside.
Pressure decreases.
Pressure in the body pushes
Liquid up. CM is moved up/
Inside the bird, there is no
Air. Just vapor of the
Liquid.



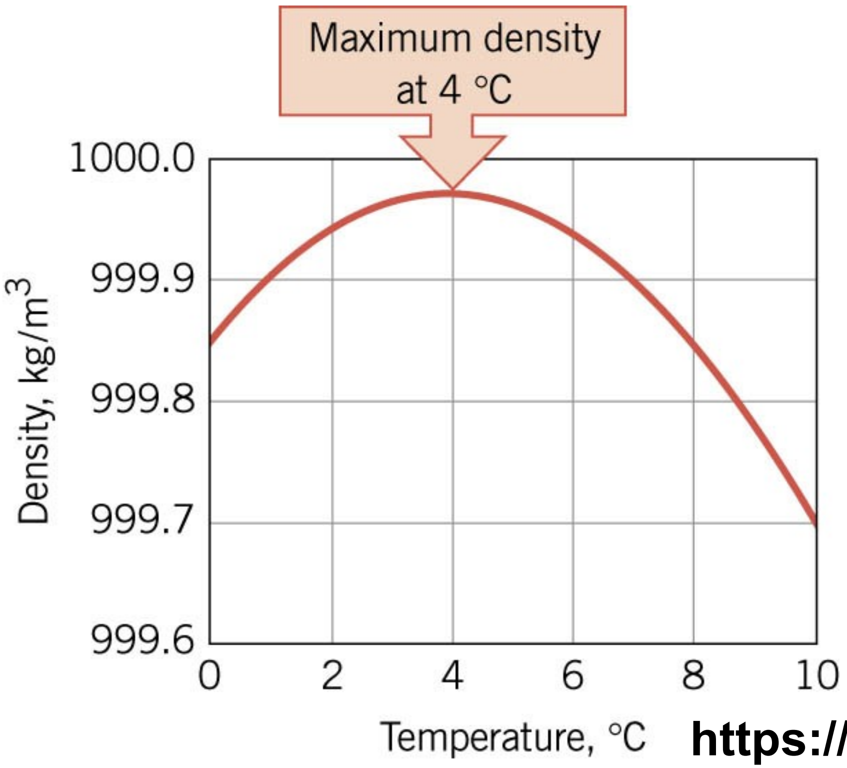
Ref: Paul Hewitt.
The steam pushed the paddles and cool down. There is condensation .
So low pressure → differential of pressure between the 2 sides of the paddles → Work is done.

Evaporation, Continued-1

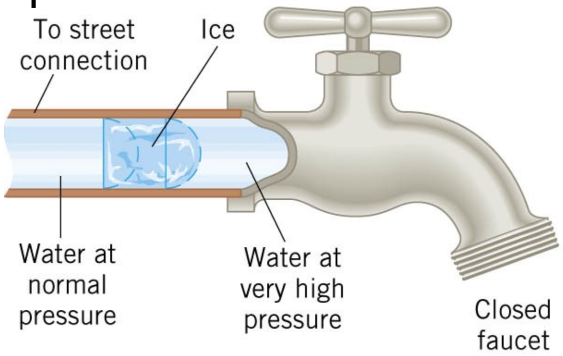
- Important in cooling our bodies when we overheat
  - Sweat glands produce perspiration.
  - Water on our skin absorbs body heat as evaporation cools the body.
  - Helps to maintain a stable body temperature.

Evaporation is a cooling body. Our warm Body loses heat to turn water vapor to liquid

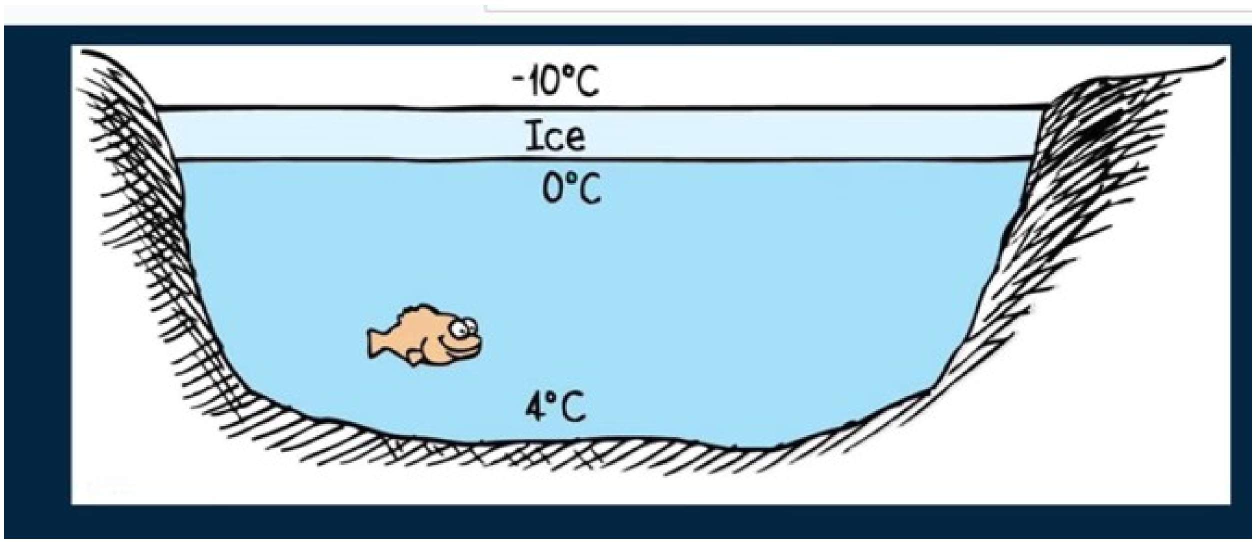
12.5 Volume Thermal Expansion



Expansion of water.

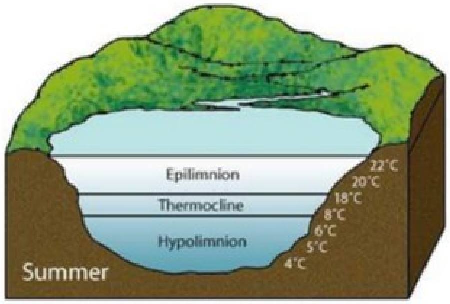


https://www.youtube.com/watch?v=u1s17LFXHqo



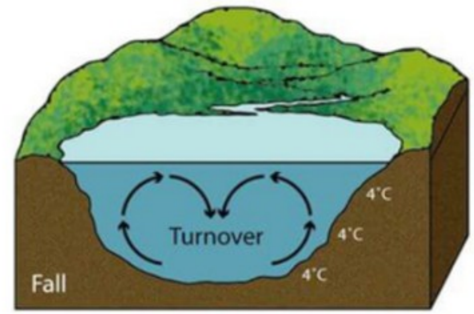
This process is crucial for redistributing nutrients and oxygen throughout the lake, which is vital for aquatic life.

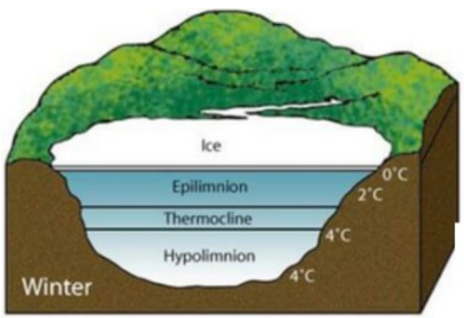
Lake turn over.



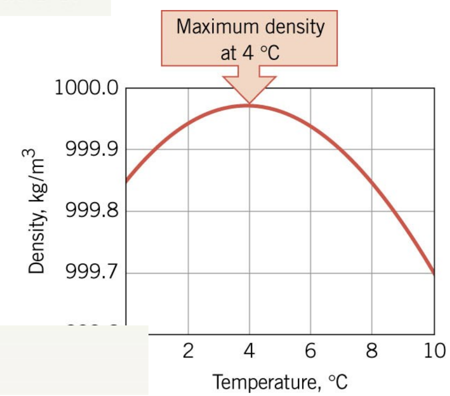
summer: The surface water is warm and less dense, sitting on top of colder, denser ter at the bottom.

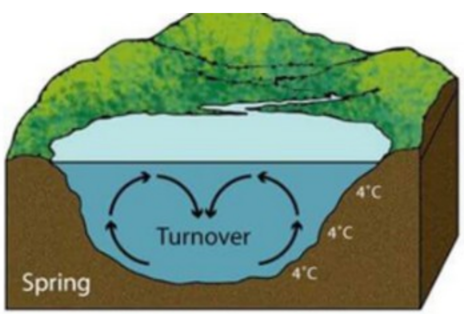
In fall: As the surface cools, it becomes denser and sinks, mixing with the deeper water. This continues until the entire lake reaches 4°C.





. In winter: As the surface water cools below 4°C, it becomes less dense again and stays on top. Eventually, it may freeze, while the bottom of the lake remains at about 4°C.

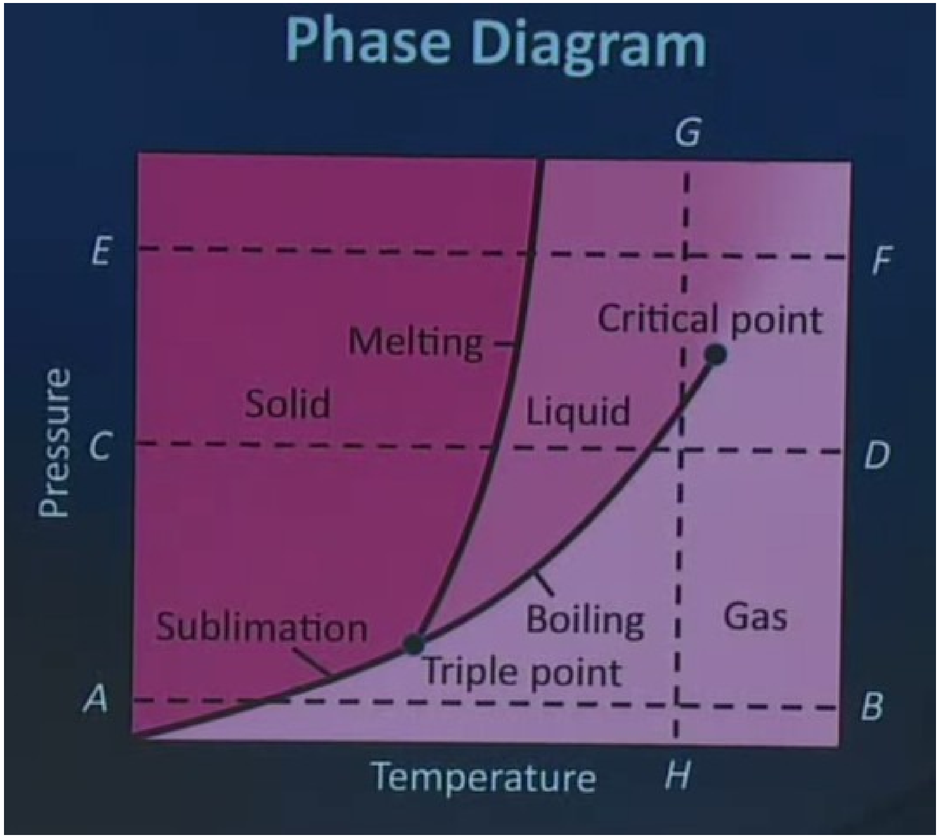




. In spring: As the surface warms to 4°C, it sinks, causing another turnover.

Phase diagrams

These diagrams are crucial in understanding how substances transition between solid, liquid, and gas phases depending on external conditions.

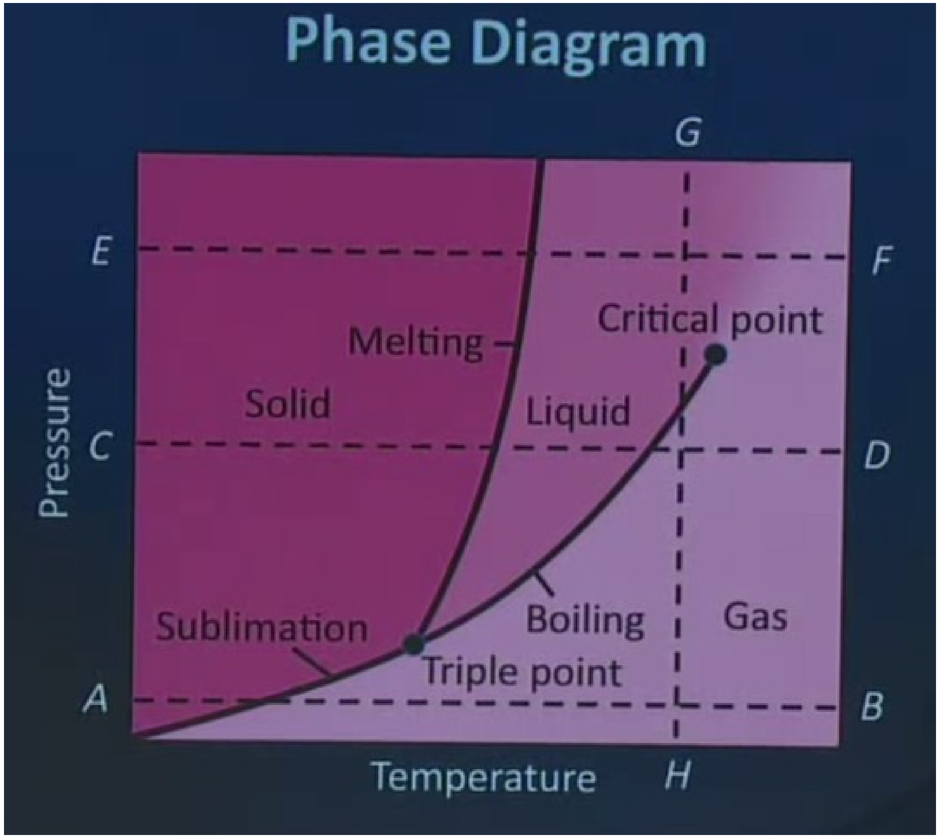


Phase Boundaries: Lines on the diagram represent boundaries between phases. These lines indicate conditions (temperature and pressure) where two phases can coexist in equilibrium.

Example: CD for a given pressure. If you increase the temperature, the matter goes from solid to liquid to gas. To increase the temperature, use the specific heat c. To change phase, use the latent heat.

AB. For dry ice. At atmospheric pressure. You go from solid to gas (sublimation).

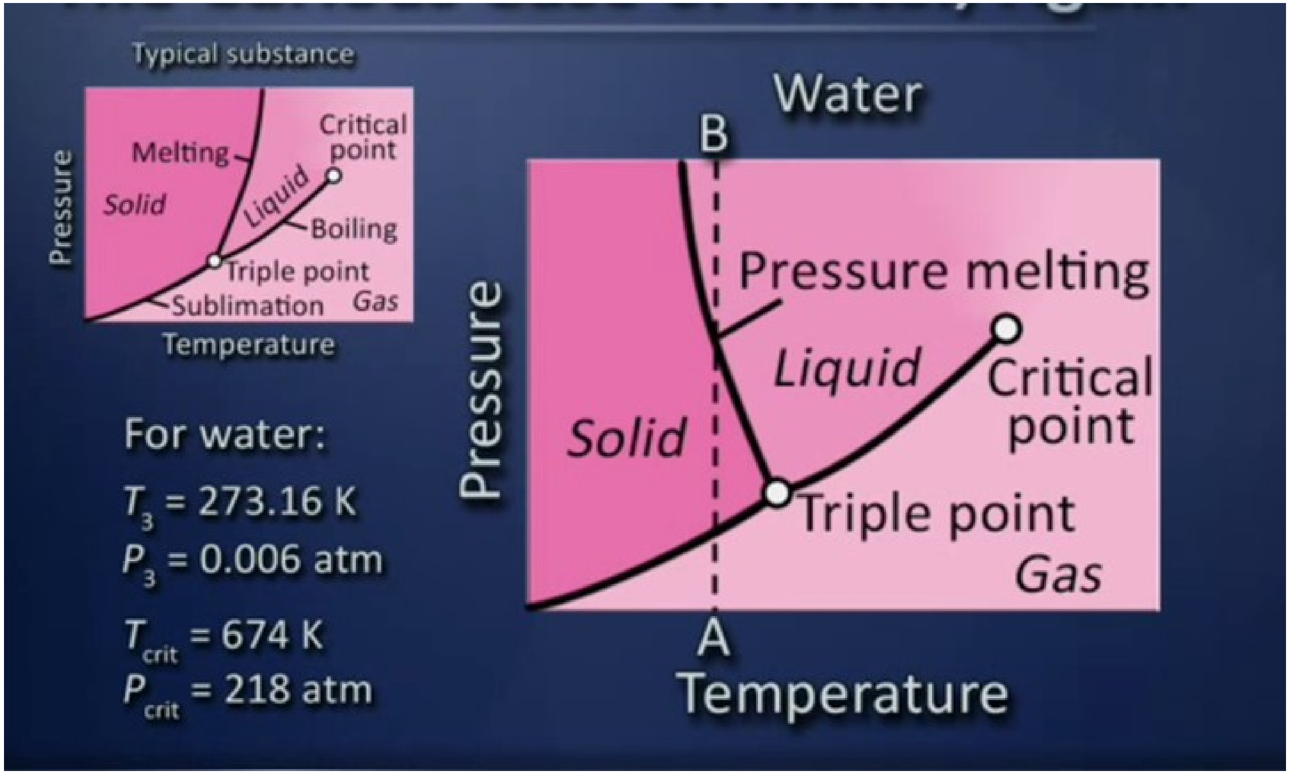
GH. for the same temperature. If you decrease The pressure, you can go from liquid to gas. Remove air from a closed water container and The water boils. On mountain top, the pressure Is less → water boils at a temperature less Than 100C.



Triple Point: The point on the diagram where all three phases (solid, liquid, gas) coexist in equilibrium. It represents the specific temperature and pressure conditions where the substance can exist simultaneously in all three phases.
Water: 273.16 kelvins (0.01 °C) and a pressure of 611.73 pascals (ca. 6.1173 millibars, 0.0060373057 atm).

Critical Point: The point beyond which the distinction between liquid and gas phases becomes less meaningful (super critical fluid). At this point, the substance behaves like a dense gas and liquid.

Water is “weird”



Water is curious. See boundary solid/liquid. The line is curved to the left.
This is because going from solid to liquid, between 0 and 4C, the density increases.

So for a given temperature. Along AB. If you increase the temperature, you can go
From solid to liquid. For another substance, its the opposite.

Summary

When heat flows into a material, it doesn't just warm up

- It may undergo thermal expansion
  - Or contraction for H_2O between 0° and 4°C
- It may undergo a phase change, with temperature remaining constant
  - Energy involved in phase change is $Q = Lm$
- Details depend on both temperature and pressure, as shown in phase diagrams